Ultra-Low-Power PMU for a Biomedical Sensor Interface

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1 Project Introduction

- The goal of this project is to design a low-power Power Management Unit (PMU) for a battery-powered biomedical sensor interface.The PMU supplies the power and reference signals required by the sensing system while operating from a small battery that must support a long operating lifetime.
- The design must therefore achieve a good compromise between power consumption, accuracy, noise, PSRR, regulation, startup behavior, and robustness.The PMU will interface with an Analog Front End (AFE) and a 14-bit ADC.
- The AFE itself is not part of the project; instead, the ADC and battery specifications are provided so that the students can derive the required PMU performance.
- The project should be approached as a real IC-design problem: start from the system requirements, derive the block-level specifications, choose an architecture, and then implement and verify the circuits.

2 PMU Blocks to Design

The PMU shall contain the following main blocks:

2.1. Battery and operating conditions

ReqNr	Symbol	Description	Min	Typ	Max	Unit	Remark
R.001	Vbat	Battery range	2.3	3	3.2	V	
R.002	Bcap	Battery capacity			150	mAh	Should operate for 90 days
R.003	Vdd	Analog supply	1.1	1.2	1.3	V	Analog supply. Supplies AFE and ADC

R.004	Ipd	Powerdown current.			1	nA	The current drawn from the battery pin supplying all the PMU should not exceed 1nA
R.005	Vnoise_dc	Battery disturbance at DC	-50		50	mV	Helpful in calculating PSRR
R.006	Vnoise_10khz	Battery disturbance at 10Khz	-2		2	mV	
R.007	Vnoise_100Khz	Battery disturbance at DC	-1		1	mV	
R.008	Vnoise_1Mhz	Battery disturbance at 10Khz	-0.3		0.3	mV	
R.009	Iload	Current load from the analog supply VDD	1		50	uA	

2.2. Bandgap / Reference

R.010	**Useful givens**
	The bandgap is used to generate a reference for LDO & ADC • ADC accuracy: 14 bit • ADC FS: 1V & reference voltage 1V • ADC sampling frequency: 100KS/s • ADC sensng BW: 1Hz to 10Khz *You should use these data to implement a noise budget for you reference path & Bandgap PSRR*
R.011	**Bandgap OK**
	The state of the bandgap start-up circuit is made available as the bandgap_ok signal. It is asserted when the VBG reaches at least **95% of its steady-state value**
R.012	**DFT**
	The following signals can be forced/sensed on the ATB (Analog test bus) Sensed: • Vbg • Bandgap bias current Forced: • Vbg

R.013	Characterization conditions
	• Temp: 5°C to 45°C (typ is 31°C) • Vbat range: 2.3V to 3.2V (typ is 3V)
R.014	**Operating conditions**
	• The block has a digital input signal: Enable • If Enable='1', the BG is enabled • If Enable='0' the BG is in powerdown

ReqNr	Symbol	Description	Min	Typ	Max	Unit	Remark
R.015	icon	Current consumption				uA	Use R.002 & R.009 to budget it
R.016	vbg	Vbg accuracy	0.995	1.00	1.005	V	1V±0.5%
R.017	1/PSRR_DC	Vbg 1/PSRR at DC				mV/V	Use R.005to R.008
R.018	PSRR_min	Vbg minimum PSRR from 1hz to 1Mhz	10			dB	Use R.005 to R.008
R.019	noise_1hz_10khz	Vbg integrated noise from 1Hz to 10KHz				uVrms	Use the info provided in R.010
R.020	noise_1hz	Vbg 1Hz spot noise (1/f)				uV/sqrt(Hz)	Report the numbers
R.021	noise_1khz	Vbg 1kHz spot noise (white noise)				uV/sqrt(Hz)	Report the numbers
R.022	ibias_out	Bnadgap output bias		100		nA	
R.023	t_stup	Startup time			1	ms	To 0.5% accuracy

***A buffer between the bandgap and the ADC reference input is **optional**. The student should decide whether a buffer is required based on the ADC loading, noise, accuracy, stability, and power requirements.*

*** The total start-up of the PMU (VDD is 1.2V to 1% accuracy) is 5ms. You can budget your start-up as you want. You need to mention the rational behind your decisions*

2.3. LDO

- Design an LDO to supply the analog circuitry of the sensing system.
- Characterise the LDO using **R.013**
- DFT:
    - o VDD should be sensed on ATB
    - o VDD could be forced via ATB
- Operating condition:
    - o Refer to **R.014**

ReqNr	Symbol	Description	Min	Typ	Max	Unit	Remark
R.024	icon	Current consumption				uA	Use R.002 & R.009 to budget it
R.025	vdd	LDO output voltage	1.1	1.2	1.3	V	
R.026	1/PSRR_DC	Vbg 1/PSRR at DC				mV/V	Use R.005 to R.008
R.027	PSRR_min	Vbg minimum PSRR from 1hz to 1Mhz	10			dB	Use R.005 to R.008
R.028	noise_1hz_10khz	Vbg integrated noise from 1Hz to 10KHz				uVrms	Use the info provided in R.010
R.029	noise_1hz	Vbg 1Hz spot noise (1/f)				uV/sqrt(Hz)	Report the numbers
R.030	noise_1khz	Vbg 1kHz spot noise (white noise)				uV/sqrt(Hz)	Report the numbers
R.031	LR	Load regulation			1%	mV/uA	The load ranges from 1uA to 50uA
R.032	t_stup	Startup time			1	ms	To 0.5% accuracy

***You can assume that the AFE has extra supply rejection and that the anti-aliasing filter is filtering the nosie above the 50Khz (half the sampling frequency)*
*** The total start-up of the PMU (VDD is 1.2V to 1% accuracy) is 5ms. You can budget your start-up as you want. You need to mention the rational behind your decisions.*
The LDO must provide a stable regulated output from the battery over the specified operating conditions.

2.4. Bias Generator

- Generate a 100-nA bias current derived from the bandgap/reference to bias the AFE blocks.
- Refer to **R.022**

Only one bias output is required for this project.

2.5. POR / Battery Monitor

- Design a Power-On-Reset / battery-monitoring circuit that monitors the battery voltage and controls PMU startup and shutdown.

ReqNr	Symbol	Description	Min	Typ	Max	Unit	Remark
R.033	icon	Current consumption				uA	Use R.002 & R.009 to budget it
R.034	Vth_high	High threshold			2.5	V	Above this threshold, POR is '1'
R.035	Vth_low	High threshold	2.3			V	Below this threshold POR is '0'
R.036	Vhyst	PoR hysteresis			200	mV	

2.6. DFT — Design for Test

The DFT circuitry is included to allow the PMU and the rest of the chip to be **tested and debugged in the lab** after fabrication.
The DFT functionality is divided into two main categories:

- **Signals to be sensed**
- **Signals to be forced**

2.6.1. Sense

The sense function allows an internal voltage or current to be observed externally through an Analog Test **Bus (ATB)**.
For example, to observe the internal **VBG**, a digital test-enable signal (tm_bg_sense_en_vbat) activates a test switch that connects VBG to an ATB. The voltage can then be measured externally during lab testing.
The test switch must be designed such that its connection does not significantly disturb the normal operation of the circuit.

2.6.2. Force

The force function allows an internal voltage or current to be **externally overridden** for debugging and characterization.
Again, a digital test-enable signal (*tm_bg_force_en_vbat*) activates the corresponding test switch. An external voltage or current can then be applied through the ATB to override the internal signal.
For example, if the bandgap does not operate correctly, the designer should be able to **externally force the VBG voltage.** This allows the rest of the PMU/chip to be tested independently of the bandgap and helps identify where a system-level problem originates.
The same concept can be applied to other internal voltages or currents, such as the **LDO output and bias current.**

2.6.3. DFT Requirements for This Project

The PMU shall provide DFT access to the following signals:
Sense:

- VBG
- LDO output voltage
- 100-nA bias current

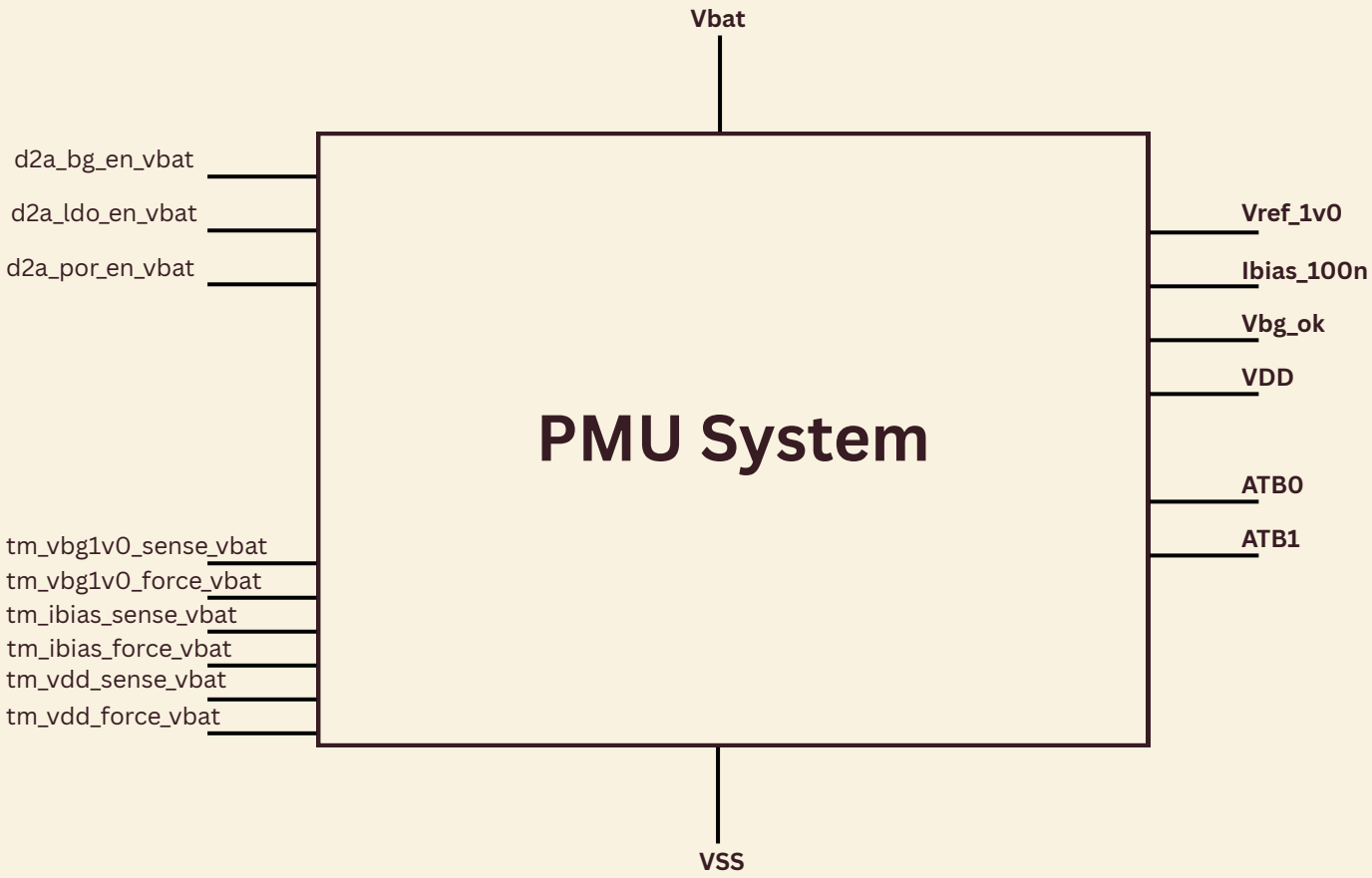
Force:

- VBG
- LDO output voltage
- 100-nA bias current

The project provides **two ATBs, ATB0 and ATB1**. You are free to distribute the DFT functions between the two ATBs according to your chosen implementation.
Your DFT implementation should consider:

- Test-switch size and ON resistance
- Leakage
- Loading of the internal circuit
- Impact on normal operation
- Required test-enable signals
- Ability to isolate the internal circuit when a signal is being forced
    - If you force Vbg, BG should be disbaled.

3 Checkpoint 1: PMU Architecture and Budgeting (15 Points)



- Implement the PMU block diagram based on the requirements and Figure 1.
- Define the current consumption budget for each block.
- Allocate the PSRR and noise budget across all blocks.
- Add any required digital control signals (e.g., enable, trim code, test mode).
- Show all calculations used to derive the specifications.
- Justify the chosen architecture and verify that it is realistic and capable of meeting all requirements

4 Checkpoint 2: Bandgap Reference (BG) Design (15 Points)

- Design the transistor-level Bandgap Reference.
- Define the VBG equation and the VBG error equation.
- Calculate the worst-case VBG accuracy, including all error sources:
    - o Temperature drift
    - o Supply variation
    - o Offset
    - o Mismatch
    - o Trim error (if applicable)
- Perform PVT and Monte Carlo simulations.
- Include:
    - o Verification plan table
    - o Key simulation results
    - o Topology justification
    - o Design challenges and trade-offs

5 Checkpoint 3: Bias Generator Design (100 nA) (15 Points)

- Design the transistor-level bias circuit generating 100 nA.
- Verify accuracy across PVT and Monte Carlo simulations.
- Include:
    - o Verification plan table
    - o Key simulation results
    - o Design challenges and trade-offs

6 Checkpoint 4: LDO Design (15 Points)

- Design the transistor-level LDO.
- Verify:
    - o Output voltage accuracy
    - o Load regulation
    - o Line regulation
    - o PSRR
    - o Stability
    - o Quiescent current
- Perform PVT and Monte Carlo simulations where applicable.
- Include:
    - o Verification plan table
    - o Key simulation results
    - o Design challenges and trade-offs

7 Checkpoint 5: Power-On Reset (PoR) Design (10 Points)

- Design the transistor-level PoR circuit.
- Verify functionality across PVT and Monte Carlo simulations.
- Include:
    - o Verification plan table
    - o Key simulation results
    - o Design challenges and trade-offs

8 Checkpoint 6: Test Switch and DFT Implementation (10 Points)

- Design the test switch.
- Implement DFT features in all PMU blocks.
- Verify DFT functionality at the typical corner.
- Include:
    - o Test switch Ron and leakage simulations across PVT
    - o DFT verification plan
    - o Waveforms demonstrating DFT operation
        For example, in the bandgap transient TB, activate the DFT feature and show me that you can sense/force the VBG. Repeat for eevry block that has a DFT feature.
        Do it at typical corner only.
    - o Key simulation results/plots

9 Checkpoint 7: Top-Level PMU Verification (20 Points)

- Perform transient simulations at TT, FF, and SS corners.
- Verify:
    - o Startup sequence
    - o VBG settling
    - o 100 nA bias current settling
    - o LDO output settling
    - o PoR behavior
    - o VBG_OK signal
- After startup:
    - o Apply a 1 µA load.
    - o Slowly ramp the load to 50 µA.
    - o Verify regulation and stability.
- Shutdown the PMU and verify:
    - o Correct power-down behavior of all blocks
    - o No unintended battery current consumption
- Include key waveforms:
    - o VBG
    - o IBIAS
    - o VDD
    - o PoR output
    - o VBG_OK
    - o Load current
    - o Battery current
- Provide a final summary table comparing all specifications against simulation results.

10 General Report Requirements (10 Bonus if followed)

- The report should be clear, concise, and well organized.
- Include all key plots, calculations, and simulation results required to justify the design decisions and demonstrate compliance with the specifications.
- Use **hierarchical schematics**. Avoid flat schematics whenever possible.
- Every schematic must be clearly presented and easy to read.
- Use **meaningful device names** (avoid generic names such as M1, M2, M3 without documentation).
- Create **meaningful symbols** that reflect the function of each block.
- Use **descriptive pin names** so that the purpose of each pin is immediately clear to the reader (e.g.,VBG_OK, LDO_EN, IBIAS_OUT, POR_OUT, TEST_MODE, TRIM<3:0>).
- Maintain a clean schematic layout with proper alignment and signal flow.
- Clearly label all critical signals, bias nodes, references, and control paths.
- Every verification result should be traceable to a corresponding specification.
- Include sufficient comments and annotations so that another designer can understand and review the design without additional explanations. (e.g. annotations for current value in each branch)

11 Layout floorplan (optional) (10 points bonus)

- Try to do a floorplan for your PMU, and estimate the area (optional)